

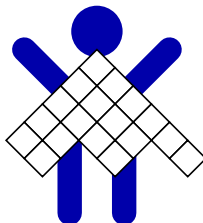
Self-conjugate core partitions: It's storytime!

Christopher R. H. Hanusa
Queens College, CUNY

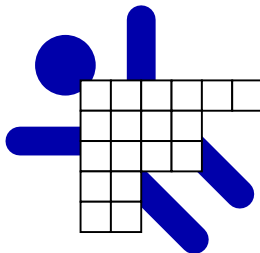
Joint work with Rishi Nath, York College, CUNY

people.qc.cuny.edu/chanusa > Talks

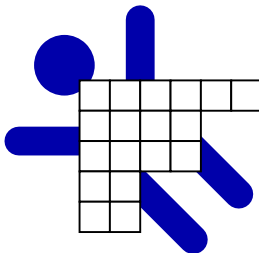
Meet Mr. Core Partition



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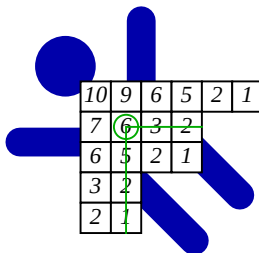


Meet Mr. Core Partition



The **Young diagram** of $\lambda = (\lambda_1, \dots, \lambda_k)$ has λ_i boxes in row i .

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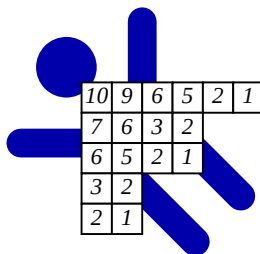


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The **hook length** of a box = # boxes below + # boxes to right + box

λ is a **t -core** if no boxes have hook length t .

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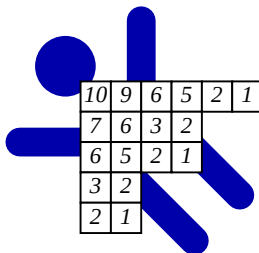
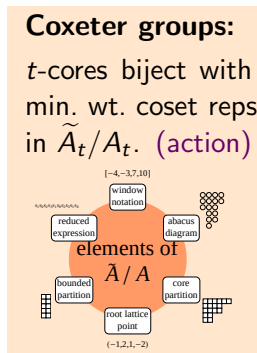
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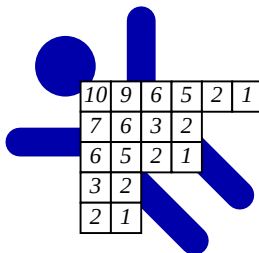
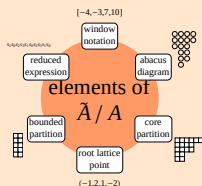
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Coxeter groups:

t -cores biject with
min. wt. coset reps
in $\tilde{A}_t/A_t$. (action)



Representation Theory:

t -cores label the
 t -blocks of
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Mock theta functions

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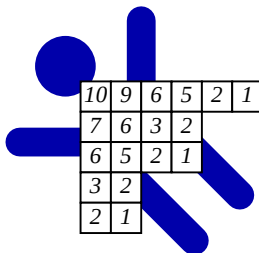
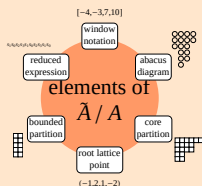
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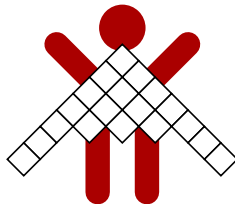
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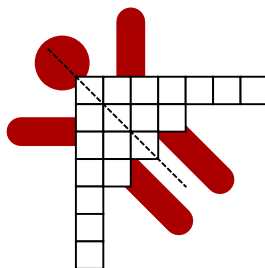
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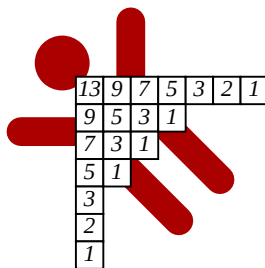
Meet Mrs. Core Partition



A partition is **self-conjugate** if it is symmetric about its main diagonal.

In this talk: Understanding self-conjugate core partitions.

Meet Mrs. Core Partition



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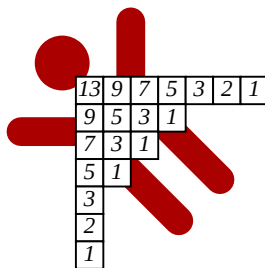
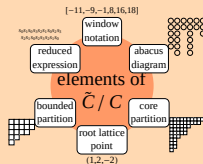
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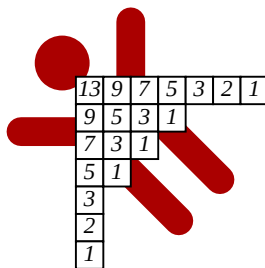
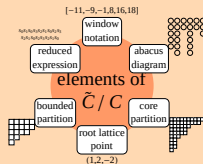
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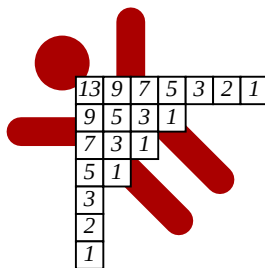
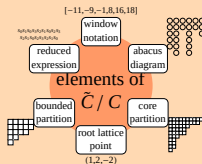
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Beauty contest



Core partitions

**Self-conjugate
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Core partitions

Generating function:

(Olsson, 1976)

$$\sum_{n \geq 0} c_t(n) q^n = \prod_{n \geq 1} \frac{(1 - q^{nt})^t}{1 - q^n}$$

Self-conjugate core partitions

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Positivity. (Granville, Ono, '96)

$c_t(n) > 0$ when $t \geq 4$.

Monotonicity. (Stanton '99)

Conjecture: $c_{t+1}(n) \geq c_t(n)$

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Positivity? ✓ (Baldwin et al, '06)

$sc_t(n) > 0$ for $t = 8, \geq 10, n > 2$.

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Monotonicity?

What else can we say?

Understanding Monotonicity

Self-conjugate partitions of 22

									Total
6-core	×	×	✓	×	×	✓	×	×	2
7-core	×	×	×	×	✓	×	×	×	1
8-core	×	✓	×	✓	✓	×	✓	×	4
9-core	×	×	✓	✓	×	×	×	×	2
10-core	✓	✓	✓	✓	✓	✓	✓	✓	8
11-core	×	×	×	×	×	✓	×	✓	2
12-core	✓	✓	✓	✓	✓	✓	✓	✓	8
13-core	✓	✓	✓	✓	×	×	✓	✓	6
14-core	✓	✓	✓	✓	✓	✓	✓	✓	8
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Understanding Monotonicity

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► **Much variability!**

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► **Most** partitions are t -cores (t large)

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- **Much variability!**
- **Most** partitions are t -cores (t large)
- Self-conjugate cores do not satisfy $sc_{t+1}(n) \geq sc_t(n)$.

Understanding Monotonicity

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- **Much variability!**
- Self-conjugate cores do not satisfy $sc_{t+1}(n) \geq sc_t(n)$.

- **Most** partitions are t -cores (t large)
- Self-conjugate cores might satisfy $sc_{t+2}(n) \geq sc_t(n)$.

Monotonicity Conjectures & Theorems



Monotonicity Conjecture. (Stanton '99)

$$c_{t+1}(n) \geq c_t(n) \text{ when } 4 \leq t \leq n-1.$$

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Even Monotonicity Conjecture. (Hanusa, Nath '12)

$$sc_{2t+2}(n) > sc_{2t}(n) \text{ for all } n \geq 20 \text{ and } 6 \leq 2t \leq 2\lfloor n/4 \rfloor - 4$$

Odd Monotonicity Conjecture.

$$sc_{2t+3}(n) > sc_{2t+1}(n) \text{ for all } n \geq 56 \text{ and } 9 \leq 2t+1 \leq n-17$$

Monotonicity Conjectures & Theorems



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Odd Monotonicity Conjecture.

$sc_{2t+3}(n) > sc_{2t+1}(n)$ for all $n \geq 56$ and $9 \leq 2t+1 \leq n-17$

Some progress:

Theorem. $sc_{2t+2}(n) > sc_{2t}(n)$ when $n/4 < 2t \leq 2\lfloor n/4 \rfloor - 4$.

And: $sc_{2t+3}(n) > sc_{2t+1}(n)$ for all $n \geq 48$ and $n/3 \leq 2t+1 \leq n-17$.

Key idea: The t -quotient of λ

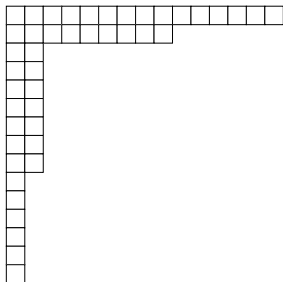


We can define the t -core λ^0 of any partition λ . Successively remove hooks of hooklength t and keep track in λ 's t -quotient.

Key idea: The t -quotient of λ



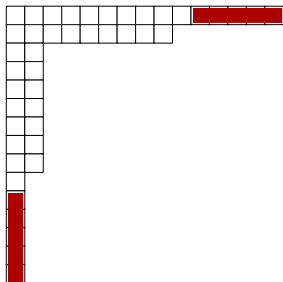
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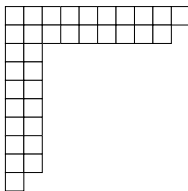


$$(\square, \emptyset, \emptyset, \emptyset, \square)$$

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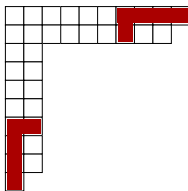


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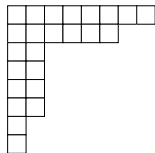


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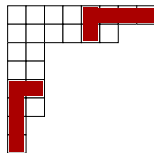


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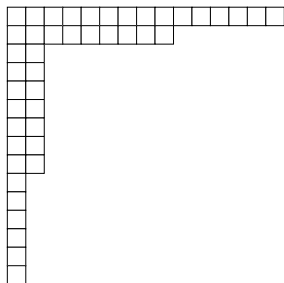


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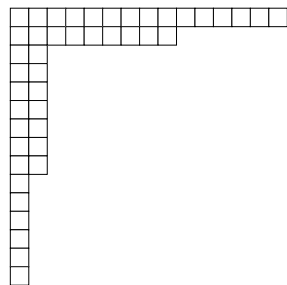
5-core

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A self-conjugate
non- t -core
partition of n

unique
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5-core

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5-quotient

A “symmetric” list of t partitions
(i boxes total)
+
a self-conj t -core partition of $n - i \cdot t$

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Proposition. For $n/3 < 2t + 1 \leq n/2$,
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Consequence: For $n/4 < 2t \leq n/2$,

$$tsc(n - 4t - 4) > (t + 1)sc(n - 4t).$$

$$sc_{2t+2}(n) > sc_{2t}(n) \quad \longleftrightarrow$$

or instead

$$\frac{sc(n - 4t - 4)}{sc(n - 4)} \leq \frac{t}{t + 1}.$$

Look Ma, No cores!

Positivity for small t

We found some holes in the literature:

$sc_2(n) = 0$ **except when** n triangular.

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Sums of squares

Theorem. If $n = (8m + 1)4^k - 2$ for $m, k > 0$, then $sc_7(n) = 0$.

Proof. (Garvan, Kim, Stanton '90) shows that

$$sc_7(n) = \# \text{ triples } (x_1, x_2, x_3) \text{ satisfying} \\ n = 7x_1^2 + 2x_1 + 7x_2^2 + 4x_2 + 7x_3^2 + 6x_3$$

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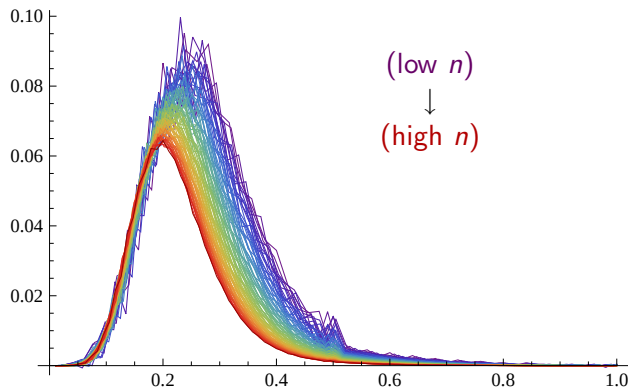
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Here: The only integers NOT sum
of 3 squares of diff. residues mod 7:
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Unimodality and Asymptotics

We conjecture $sc_{t+2}(n) > sc_t(n)$; structure of increase?

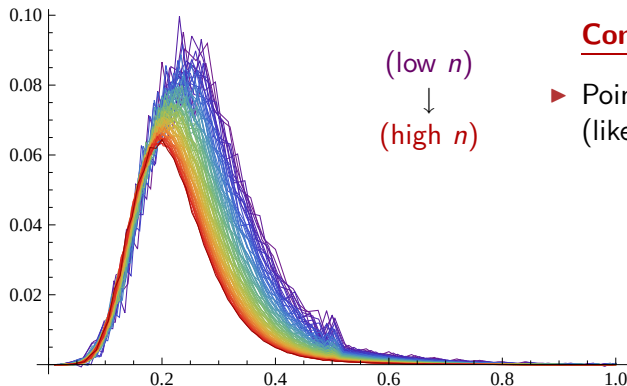
Plot Normalized increase for different n : $(sc_{t+2}(n) - sc_t(n)) / sc(n)$



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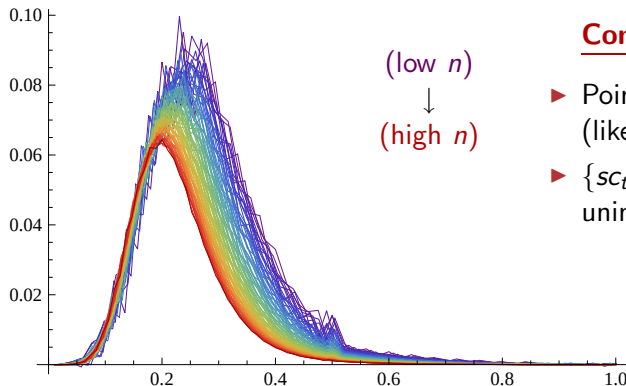
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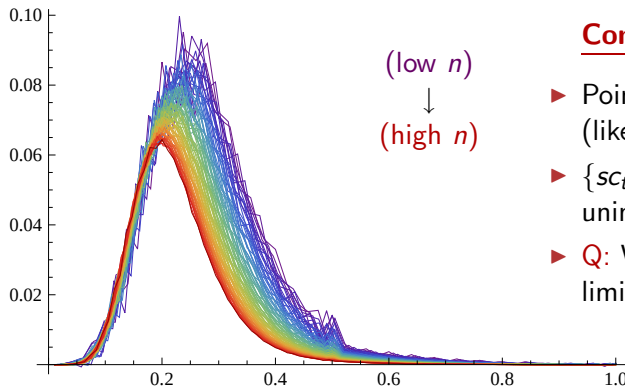
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- ▶ Q: What is the
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Other peculiarities

Conjecture: There are infinitely many n such that $sc_9(n) < sc_7(n)$.

Includes many (but not all) values of $n \equiv 82 \pmod{128}$:

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Conjecture: For $n \geq 0$, $sc_7(4n + 6) = sc_7(n)$.

Conjecture: Let n be a non-negative integer.

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2. Suppose $n \geq 1$. Then $sc_9(4n + 1) > 1.9 sc_9(n)$.
3. Suppose $n \geq 17$. Then $sc_9(4n + 3) > 1.9 sc_9(n)$.
4. Suppose $n \geq 1$. Then $sc_9(4n + 4) > 2.6 sc_9(n)$.

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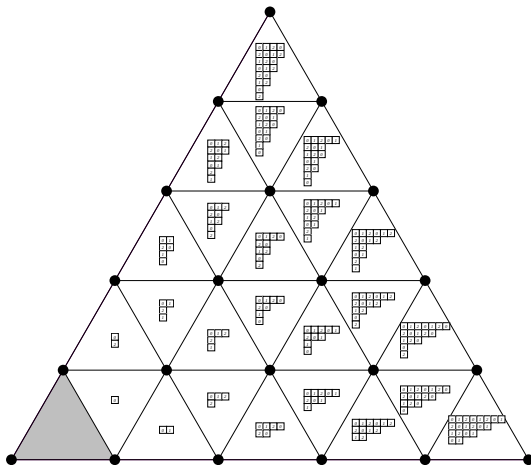
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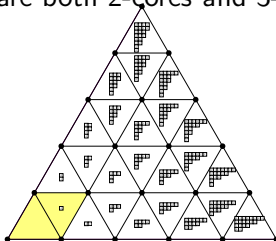
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 - ▶ Geometrical interpretation of cores:

The bijection between 3-cores and alcoves



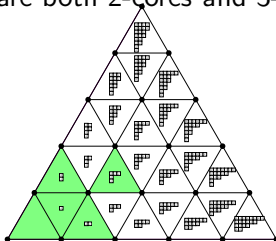
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How many partitions are both 2-cores and 3-cores? **2**.



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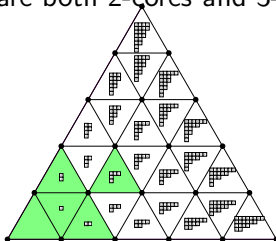
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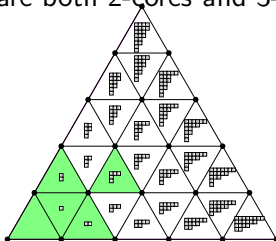


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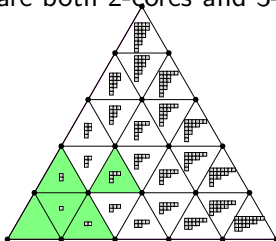
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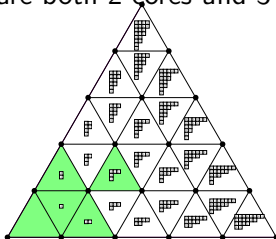
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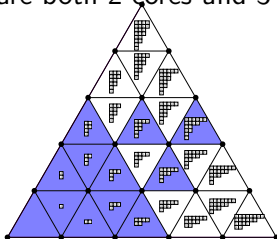
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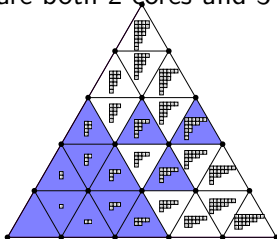
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Fishel–Vazirani proved an alcove interpretation of $n/(mn+1)$ -cores. ↻ 🔍

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- ▶ **Question:** What is the average size of an s/t -core partition?
 - ▶ In progress (on pause).
We “know” the answer, but we have to prove it!
 - ▶ Working with Drew Armstrong, University of Miami.

Thank you!

Slides available: people.qc.cuny.edu/chanusa > **Talks**

Interact: people.qc.cuny.edu/chanusa > **Animations**



Gordon James and Adalbert Kerber.

The representation theory of the symmetric group,
Addison-Wesley, 1981.



Christopher R. H. Hanusa and Rishi Nath.

The number of self-conjugate core partitions. [arXiv:1201.6629](https://arxiv.org/abs/1201.6629)



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Abacus models for parabolic quotients of affine Coxeter groups
Journal of Algebra. Vol. 361, 134–162. (2012) [arXiv:1105.5333](https://arxiv.org/abs/1105.5333)